

Souradeep Bhattacharya

PHD CANDIDATE · DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

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Research Interests

Cybersecurity for cyber-physical systems with a focus on critical infrastructure and smart grid technologies. Research interests include the application of explainable AI/ML techniques for anomaly detection, physics-informed approaches for resilient attack mitigation, security of ICS/OT communication protocols, zero trust for ICS/OT systems, attack surface analysis and threat modeling, and real-time digital simulation with hardware-in-the-loop (HIL) experimentation for CPS.

Education

Iowa State University

Ames, Iowa, USA

PH.D. COMPUTER ENGINEERING

January 2022 - Expected 2026

Advisor: Prof. Manimaran Govindarasu

GPA: 3.87/4

Thesis: AI-driven Algorithms Zero Trust Security Assessment, and Cyber Attack Detection and Mitigation for Smart Grids.

Iowa State University

Ames, Iowa, USA

M.S. COMPUTER ENGINEERING

January 2022 - May 2025

Advisor: Prof. Manimaran Govindarasu

GPA: 3.74/4

Thesis: CAN-ADS: Machine Learning-based Anomaly Detection System for CAN Bus Network in Agriculture Machinery.

Maulana Abul Kalam Azad University of Technology

Kolkata, West Bengal, India

B.TECH APPLIED ELECTRONICS AND INSTRUMENTATION ENGINEERING

July 2012 - June 2016

Project Advisor: Prof. Priyadarshiny Dhar

GPA: 8.39/10

Final Year Project: Dual-Tone Multi-Frequency (DTMF) Controlled Robotic Vehicle.

Research Experience

Graduate Research Assistant

Ames, Iowa, USA

Department of Electrical and Computer Engineering, Iowa State University

January 2022 - present

- Research Assistant in the PowerCyber Laboratory at ISU with Prof. Manimaran Govindarasu.
- Contributed to three US DOE CESER-funded projects, focusing on enhancing cybersecurity and resilience in solid state transformer (SST)-based substations and Distributed Energy Resources (DER) & microgrid-integrated grid networks.
- Developed end-to-end AI pipelines (dataset generation, pre-processing, and analysis to model training, validation, and deployment) and optimized ML models for edge deployment (NVIDIA Jetson, Raspberry Pi), enabling efficient, low-latency inference for cyber-attack detection and mitigation in resource-constrained environments.
- Developed and deployed real-time ML-based anomaly detection systems for ICS/OT protocols (DNP3, IEC 61850, Modbus, CANbus) with high accuracy and low latency.
- Developed a novel incremental ML-based anomaly detection system (ADS) and physics-informed neural network (PINN)-based attack mitigation for SST systems, enabling online learning for the real-time detection and mitigation of evolving attack patterns in dynamic power system environments.
- Developed a real-time, cloud-based cyber situational awareness platform tailored for DER networks, combining edge ML detection, Knowledge Graph-based correlation, and Explainable AI (XAI) to generate high-confidence alerts provide real-time correlation of potential threats to enhance overall security against cyber-attacks.
- Developed Zero Trust Maturity Model Framework and AI-driven self-assessment tool, tailored for smart grid, leveraging large language models (LLMs) and Retrieval-Augmented Generation (RAG) algorithms, to guide the evaluation and adoption of zero trust solutions within grid environments.
- Conducted power system simulations and cyber-attack-defense orchestration on power grid models using hardware-in-the-loop (HIL) testbeds with real-time digital simulators (OPAL-RT), industry-grade devices, platforms, and software tools, including SEL Relays, Siemens PowerTG, SICAM PAS for SCADA/RTU, NVIDIA Jetson AGX, and Raspberry Pi.
- Contributed to the development and writing of research proposals for US DOE and NSF funding opportunities.

Teaching Experience

Iowa State University

Ames, Iowa, USA

TEACHING ASSISTANT

January 2023 - Present

Course: CPRE 5390: Cyber-Physical System Security for the Smart Grid

Academic Sessions: Spring 2026, Spring 2025, Spring 2024, Spring 2023

Responsibilities:

- Assisted students with laboratory experiments.
- Assisted in the development of lecture materials and course content.
- Graded assignments and exams to evaluate student performance and provide constructive feedback.
- Delivered lectures on topics including CPS Security, SCADA, Cybersecurity Testbeds, Grid Communication Protocols, Intrusion and Anomaly Detection Systems, Distributed Energy Resources, EV Charging Infrastructures, and Zero Trust.

Professional Experience

National Laboratory of the Rockies (formerly NREL)

May 2025 - August 2025

Cybersecurity Research Intern

Golden, Colorado, USA

- Developed of lab-based demonstrations using NREL Cyber Range to assess and validate Cyber-Informed Engineering (CIE)-based mitigation strategies for a realistic cyber-attack use cases on energy infrastructure.
- Contributed to the development of CIE Photovoltaic Analysis Tool focused on supporting utilities to determine CIE-based protections for a PV installation.

MidAmerican Energy Company (MEC)

May 2024 - August 2024

Cybersecurity Operations Intern

Des Moines, Iowa, USA

- Deployed Dragos platform for OT/SCADA systems, configuring sensors, identifying critical assets, and creating custom threat detection rules.
- Monitored SIEM systems, analyzed anomalies, and optimized alerts to enhance incident detection and response.
- Coordinated with cross-functional teams to ensure effective deployment of cybersecurity solutions.

Johnson Controls, Inc. (JCI)

February 2021 - December 2021

Project Lead

Pune, Maharashtra, India

- Managed and delivered multiple projects for JCI branches in New York, New Jersey, Maryland, & Massachusetts.
- Successfully implemented optimized design strategy to increase energy savings and reduce cycle-time cost.
- Mentored newly joined associates by conducting knowledge transfer sessions on project functionalities.

Senior Application Engineer

April 2018 - February 2021

Baltimore, Maryland, USA

- Performed on-site programming, installation, commissioning, and field testing of critical building automation and management systems (BMS).
- Performed integration of JCI Metasys BMS with third-party control systems utilizing BACnet, LonWorks, MODBUS, and KNX communication protocols.
- Established and maintained coordination between clients, consultants, JCI field support team, and subcontractors to facilitate timely execution and strict adherence to project quality standards.

Application Engineer

July 2016 - April 2018

Pune, Maharashtra, India

- Developed HVAC control systems and sequence of operations for Building Automation and Management Systems.

Undergraduate Engineering Intern

January 2016 - March 2016

Pune, Maharashtra, India

- Completed a three-month internship and training on HVAC and building automation.

Calcutta Electric Supply Corporation

June 2015 - July 2015

Summer Intern

Kolkata, West Bengal, India

- Completed training at a thermal power plant, gaining hands-on understanding of electricity generation and distribution through key components such as boilers, cooling towers, generators, and transformers.

Orica Explosives Limited

Winter Intern

January 2015 - February 2015

Gomia, Jharkhand, India

- Completed training in commercial explosives manufacturing for the mining industry, gaining exposure to field applications of devices such as motors, generators, flowmeters, tachometers, sound level meters, and variable frequency drives.

Damodar Valley Corporation

Summer Intern

May 2014 - July 2014

Mejia, West Bengal, India

- Completed training at a thermal power plant, gaining extensive knowledge of electricity generation along with practical exposure to process control and industrial instrumentation.

Publications

JOURNALS

- S. Bhattacharya** and M. Govindarasu, "Physics-Informed Neural Network-Based Detection and Mitigation of Side-Channel Noise Intrusion in Solid State Transformers," in *IEEE Transactions on Power Electronics* (Under Review).
- S. Bhattacharya**, M. D. R. Greidanus, S. Gupta, D. Sur, S. K. Mazumder and M. Govindarasu, "Machine-Learning-Based Adaptive Anomaly Detection for Control Feedback Interferences in Solid-State Transformers," in *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, vol. 6, no. 4, pp. 1840-1852, Oct. 2025, doi: 10.1109/JESTIE.2025.3589164.
- N. Saqib, **S. Bhattacharya**, K. K. Challa, and M. Govindarasu, "Attack resilient microgrid: Impact characterisation, online anomaly detection and mitigation," *IET Smart Grid*, vol. 9, no. 1, p. e70070, 2026, doi: 10.1049/stg2.70070.
- D. Popoola, **S. Bhattacharya**, A. Hahn, and M. Govindarasu, "AI-enabled Security Operation Center (SOC) for Modern Power Grids," in *IEEE Power & Energy Society Magazine* (Under Review).

BOOK CHAPTERS

- S. Bhattacharya**, N. Saqib, and M. Govindarasu, "Anomaly Detection, Cyber Situational Awareness, and Attack Mitigation for DER Monitoring and Control," Book Chapter in *Cybersecurity and Resilience of Modern Power Systems*, Wiley-IEEE Press (Accepted).

PEER-REVIEWED CONFERENCES

Published/Accepted

- S. Bhattacharya** and M. Govindarasu, "Real-time Cybersecurity Situational Awareness Framework for Agriculture Machinery-based IoT Networks," *2025 34th International Conference on Computer Communications and Networks (ICCCN)*, Tokyo, Japan, 2025, pp. 1-9, doi: 10.1109/ICCCN65249.2025.11133877.
- S. Bhattacharya**, R. G. Gallolukankanamalage, B. L. Steward, and M. Govindarasu, "ML-based Anomaly Detection for CAN Bus Network in Agriculture Machinery," *Proceedings of the AAAI Symposium Series*, vol. 4, no. 1, pp. 416-423, Nov. 2024. DOI: 10.1609/aaaiss.v4i1.31826.
- S. Bhattacharya**, M. D. Roig Greidanus, S. Gupta, D. Sur, S. K. Mazumder and M. Govindarasu, "Machine-Learning-based Anomaly Detection System for Cyber-Physical Attack on a Solid-State Transformer," *IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society*, Chicago, IL, USA, 2024, pp. 1-6, doi: 10.1109/IECON55916.2024.10905322.
- S. Bhattacharya**, N. Saqib, and M. Govindarasu, "ML-based Anomaly Detection System for IEC 61850 Communication in Substations," *2024 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1-5, 2024. DOI: 10.1109/PESGM51994.2024.10688773.
- S. Bhattacharya**, M. Govindarasu, M. Girdhar, J. Hong, "A Quantitative Methodology for Attack-Defense Analysis of EV Charging Infrastructure," *2023 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1-5, 2023. DOI: 10.1109/PESGM52003.2023.10252252.
- S. Bhattacharya**, B. Hyder, M. Govindarasu, "ICS-CTM2: Industrial Control System Cybersecurity Testbed Maturity Model," *2022 Resilience Week (RWS)*, pp. 1-6, 2022. DOI: 10.1109/RWS55399.2022.9984023.

7. D. Popoola, **S. Bhattacharya**, and M. Govindarasu, "Self-Verifying Anomaly Detection using Explainable AI for Cybersecurity of DER Networks," in *2026 IEEE Power & Energy Society General Meeting (PESGM)*, Montreal, Quebec, Canada, 2026 (Accepted).
8. H. Ravichandran, **S. Bhattacharya**, N. Saqib, and M. Govindarasu, "Real-time Edge-based Protocol Converter and Testbed-based Evaluation for DER Networks," in *2026 IEEE Power & Energy Society General Meeting (PESGM)*, Montreal, Quebec, Canada, 2026 (Accepted).
9. D. Popoola, **S. Bhattacharya**, and M. Govindarasu, "CySIDER: Cybersecurity Situational Intelligence Framework for DER Networks," *2025 Resilience Week (RWS)*, National Harbor, MD, USA, 2025, pp. 1-10, doi: 10.1109/RWS66711.2025.11304446.
10. N. Saqib, F. C. Joseph, K. K. Challa, **S. Bhattacharya** and M. Govindarasu, "Impact of False Data Injection Attack on DER Integrated Distribution Systems," *2025 IEEE Power & Energy Society General Meeting (PESGM)*, Austin, TX, USA, 2025, pp. 1-5, doi: 10.1109/PESGM52009.2025.11225774.
11. S. Bose, **S. Bhattacharyaa**, N. Saqib and M. Govindarasu, "ZT-MTD: Zero Trust-enabled Dynamic Moving Target Defense for DER Networks," *2025 IEEE Power & Energy Society General Meeting (PESGM)*, Austin, TX, USA, 2025, pp. 1-5, doi: 10.1109/PESGM52009.2025.11224995.
12. N. Saqib, **S. Bhattacharya**, B. Hyder and M. Govindarasu, "Cyber Attack Impact Characterization for IEC 61850-based Substations," *IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society*, Chicago, IL, USA, 2024, pp. 1-6, doi: 10.1109/IECON55916.2024.10905837.
13. N. Saqib, **S. Bhattacharya**, K. Challa, and M. Govindarasu, "Impact Characterization of Cyber Attacks on Grid-Connected Microgrid Operations," *2024 56th North American Power Symposium (NAPS)*, pp. 1–6, 2024. DOI: 10.1109/NAPS61145.2024.10741631.
14. Shantanu Gupta, Mateo D Roig Greidanus, Silvanus A D'silva, **S. Bhattacharya**, Sudip K Mazumder, M Shadmand, Manimaran Govindarasu, Taesic Kim, Juan C Balda, Xiaoqing Song, Rambabu Adapa, Mohammad Shahidehpour, "Cyber Resiliency of a Solid-State Power Substation," *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, pp. 2293–2300, 2024. DOI: 10.1109/APEC48139.2024.10509048.

Under Review

1. A.B. Chandrasekaran, D. Popoola, **S. Bhattacharya**, and M. Govindarasu, "ZT-DAC: ML-Driven Dynamic Access Control Policy for Zero Trust in DER Networks," in *2026 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, College Station, TX, USA, 2026 (Under Review).

In Preparation

1. **S. Bhattacharya** and M. Govindarasu, "Zero Trust Cybersecurity Maturity Model and AI-driven Self-Assessment Tool for Smart Grid," in *2026 IEEE Resilience Week* (To be submitted).
2. **S. Bhattacharya**, D. Popoola, N. Saqib, and M. Govindarasu, "AI-enabled Real-Time Cyber-Physical Situational Awareness Framework for DER Networks," in *2026 IEEE North America Power Symposium (NAPS)* (To be submitted).
3. M.M. Almomani, **S. Bhattacharya**, L.E.G. Montenegro, N. Saqib, H. Villegas, V. Ajjarapu, and M. Govindarasu, "Impact of False Data Injection Attacks on DERMS Dispatch in DER-integrated Distribution Systems," in *2026 IEEE North America Power Symposium (NAPS)* (To be submitted).
4. S. Bose, **S. Bhattacharya**, and M. Govindarasu, "Zero Trust-Enabled Authentication and Integrity-Assured Encryption for Smart Grid DER Networks," in *IEEE Internet of Things Journal* (To be submitted).

Honors, Awards, & Grants

2026	1st Place, Best Poster Award , IEEE PES T&D Conference	
2025	3rd Place, Best Poster Award , ISU EPRC 2025	
2024	Best Paper Presentation Award , IEEE IECON Annual Conference	
2024	GPSS Professional Advancement Grant , Iowa State University	\$ 400
2022	Graduate Tuition Scholarship (per academic year) , Iowa State University	\$ 21,600
2021	Operational Excellence Award , Johnson Controls, Inc.	
2020	Customer Satisfaction Award , Johnson Controls, Inc.	
2019	Customer Satisfaction Award , Johnson Controls, Inc.	
2018	Customer Satisfaction Award , Johnson Controls, Inc.	
2018	Live Quality Award , Johnson Controls, Inc.	

Presentations

- IEEE PES T&D 2026.** Presentation of selected research poster titled “AI-driven Detection and Mitigation of Noise Intrusion Attacks in Solid State Transformer-Substations.”
- IEEE PES GM 2025.** Presentation of selected research paper titled “ZT-MTD: Zero Trust-enabled Dynamic Moving Target Defense for DER Networks.”
- IEEE PES GM 2025.** Presentation of Student Research Poster titled “Machine-Learning-Based Adaptive Anomaly Detection for Control Feedback Interferences in Solid-State Transformers.”
- ISU EPRC 2025.** Presentation of Student Research Poster titled “Incremental Learning-based Anomaly Detection for Cyber-Physical Attacks in Solid-State Transformers.”
- ISU EPRC visit to MISO 2025.** Presentation on “Cybersecurity and Resiliency of DER-integrated Distribution Grid” during ISU EPRC’s visit to Midcontinent Independent System Operator (MISO) at Eagan, MN.
- IEEE IECON 2024.** Presentation of selected research paper titled “Cyber Attack Impact Characterization for IEC 61850-based Substations.”
- IEEE IECON 2024.** Presentation of selected research paper titled “Machine-Learning-based Anomaly Detection System for Cyber-Physical Attack on a Solid-State Transformer.”
- AAAI Fall Symposium Series 2024.** Presentation of selected research paper titled “ML-based Anomaly Detection System for CAN Bus Networks in Agriculture Machinery.”
- IEEE PES GM 2024.** Presentation of selected research paper titled “ML-based Anomaly Detection System for IEC 61850 Communication in Substations.”
- IEEE PES GM 2024.** Presentation of Student Research Poster titled “Real-time ML-based Intrusion Detection for EM-SNI Attacks on Solid State Transformers.”
- IEEE APEC 2024.** Presentation of selected research paper titled “Cyber Resiliency of a Solid-State Power Substation.”
- EPRC Annual Meeting 2023.** Conducted a cybersecurity training workshop during ISU EPRC’s Annual Meeting.
- EPRC Annual Meeting 2023.** Presentation of Student Research Poster titled “A Reinforcement Learning based Defense Investment Strategy Optimization in Smart Grid.”
- IEEE PES GM 2023.** Presentation of selected research paper titled “A Quantitative Methodology for Attack-Defense Analysis of EV Charging Infrastructure.”
- IEEE PES GM 2023.** Presentation of Student Research Poster titled “A Reinforcement Learning based Defense Investment Strategy Optimization in Smart Grid.”

Mentoring

2023-2024 **Undergraduate Senior Design Project**, Intrusion Detection System on Automotive CAN Bus

*Iowa State
University*

2022-2023 **Undergraduate Senior Design Project**, CySecAgri: Secure IoT for Farming

*Iowa State
University*

Outreach & Professional Development

SERVICE AND OUTREACH

2026 **Iowa Science Olympiad**, Event Supervisor, Circuit Lab C

Ames, Iowa

2026 **Iowa Science Olympiad**, Volunteer, Circuit Lab B

Ames, Iowa

2015-2016 **NSEC EIE Department Alumni Association**, Technical Head

Kolkata, India

2015-2016 **NSEC Placement Committee**, Committee Member

Kolkata, India

DEVELOPMENT

AAAI Fall Symposium Series (FSS) Workshop 2024. This workshop explored advancements in AI and their applications across multiple domains, particularly in decision-making and automation. I learned about cutting-edge AI research and methodologies, which expanded my ability to apply AI-driven solutions in professional and academic projects.

EPRC Cybersecurity Workshop 2023. Focused on the cybersecurity challenges within the energy sector. It contributed to my professional development by strengthening my ability to address security vulnerabilities in critical energy systems.

PSERC Summer School 2022. This workshop provided in-depth insights into power systems engineering, research collaboration, and emerging challenges in energy infrastructure. I gained valuable knowledge on grid resilience, renewable integration, and academic-industry partnerships, enhancing my knowledge in the field.

PEER REVIEW

IEEE Power & Energy Systems (PES) General Meeting Conference.

IEEE Power & Energy Systems (PES) Magazine.

IEEE Transactions on Industrial Informatics.

IEEE Access.

Elsevier Engineering Applications of Artificial Intelligence.

PROFESSIONAL MEMBERSHIPS

IEEE Graduate Student Membership.

IEEE Young Professionals.

IEEE Power & Energy Society (PES) Membership.

Electric Power Research Center (EPRC) Membership at Iowa State University.

Power Systems Engineering Research Center Membership.

References

Dr. Manimaran Govindarasu

ANSON MARSTON DISTINGUISHED PROFESSOR AND MURRAY J. AND RUTH M. HARPOLE PROFESSOR

- *Department:* Electrical and Computer Engineering
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Dr. Venkataramana Ajjarapu

THOMAS M. WHITNEY PROFESSOR OF ELECTRICAL ENGINEERING

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Dr. Shana Moothedath

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